

Conv-ChArT: A Convolutional–Transformer Hybrid for Accurate and Compact Fiducial Corner Detection and Pose Estimation

Abstract

Fiducial-marker pose estimation underpins a wide range of localisation tasks, from camera calibration and motion capture to visual servoing, augmented reality and mobile-platform navigation. Existing detectors fail in complementary ways: classical corner-based pipelines are accurate but abstain on a large fraction of degraded frames, while learned detectors sustain throughput by emitting confidently wrong poses. We present Conv-ChArT, a convolutional–transformer hybrid for ChArUco corner detection and pose estimation. A convolutional encoder–decoder is coupled to a multi-head self-attention bottleneck with axial rotary position encoding, supplying board-scale context that a convolutional receptive field alone cannot reach across a scale envelope spanning more than a decade in apparent board size; identity is predicted by a dedicated class map, corners are localised sub-pixel by a compact confidence-gated refinement network, and a lattice-consistency homography gate makes abstention an explicit, geometrically justified outcome. We evaluate against two baselines: the standard classical ChArUco pipeline, and the established learned baseline for this task, a two-stage convolutional system that predicts corners and identities on a coarse grid, which we fine-tune on our data; as in that system’s own coarse evaluation, its second-stage refinement network is not run. Conv-ChArT establishes a new accuracy–efficiency operating point against both. Our full pipeline uses **2.3× fewer parameters** than the learned baseline, yet reduces median pose rotation error by **6.9×** and 95th-percentile rotation error by **76×**; a variant using **7.0× fewer parameters** still surpasses it on every accuracy metric. The disparity between median and tail improvement reflects the two systems’ failure modes: the baseline sustains throughput by fitting poses to inconsistent correspondences, whereas our geometric gate declines them. On corner localisation our reference model attains 0.4063 px median and 0.6733 px 95th-percentile error at 99.53% identification accuracy. We release three width tiers, characterise them over twenty degradation factors, and report all ablations against a measured per-width reproducibility floor.

1 Introduction

Planar fiducial markers remain the standard instrument for establishing metric camera pose from a single view, and underpin a wide range of localisation tasks: extrinsic and intrinsic camera calibration, multi-camera and camera-to-sensor registration, object and platform tracking, visual servoing, motion capture, and augmented-reality anchoring. The ChArUco board, a chessboard whose white squares carry ArUco markers, combines the sub-pixel localisability of chessboard saddle points with the identifiability of coded markers, and is the default choice where accuracy matters. The classical detection pipeline is a sequence of hand-designed stages: adaptive threshold, quadrilateral extraction, marker decoding and corner interpolation. Each stage is individually interpretable and collectively brittle, since a failure at any point removes the corner entirely.

Learned detectors address this brittleness by regressing corner locations directly from pixels. Deep ChArUco [5] demonstrated that a convolutional network recovers ChArUco corners in illumination regimes where the classical pipeline abstains entirely. The learned formulation, however, introduces a failure mode the classical one does not have. A classical pipeline that cannot decode a marker reports nothing; a learned detector always emits a heatmap, and a downstream solver will fit *some* pose to whatever peaks it contains. Wherever the pose is consumed automatically, whether by a calibration that is never inspected, a tracking filter or a control loop, a confidently wrong pose is categorically worse than no pose, because nothing downstream can distinguish it from a correct one.

Conv-ChArT is designed so that accuracy and abstention are obtained together, at a parameter budget compatible with embedded deployment. We make the following contributions.

- **We present** a convolutional-transformer detector in which a self-attention bottleneck with axial rotary position encoding supplies board-scale context, and identity is predicted by a dedicated class map. Its full pipeline uses 979,458 parameters against 2,245,992 for the learned baseline, and is more accurate on every metric we measure (Sec. 4.3); the bottleneck itself is ablated in Sec. 4.6 (*Attention bottleneck*).
- **We show** that supervising the corner heatmap with a one-hot target under binary cross-entropy outperforms the Gaussian-target penalty-reduced focal loss conventional in dense keypoint detection, and that the two heads are best supervised differently (Sec. 4.6, *Supervision signal*).
- **We introduce** two geometric gates: a refinement stage applied only where its own predicted confidence supports it, and a lattice-consistency homography test that converts inconsistent correspondences into an explicit refusal rather than a pose (Sec. 4.6, *Geo-*

metric gating).

- **We propose** reporting ablations against a measured per-width reproducibility floor obtained from repeated identical training runs. The smallest resolvable effect depends jointly on width and metric, varying across the widths we train by $11.6\times$ in 95th-percentile localisation and $88\times$ in identity (Sec. 4.6, *Reproducibility floor*).

2 Related Work

Fiducial markers and classical detection. Square-marker systems [1, 2, 3, 4] encode identity in a binary grid and recover pose from four corners. ChArUco boards interleave markers with a chessboard so that saddle points, which localise far more precisely than marker corners, inherit identity from neighbouring markers. The OpenCV implementation is the de facto reference and forms one of our baselines.

Learned keypoint detection. Our heatmap formulation follows the dense-prediction lineage of CornerNet [6] and CenterNet [7], in which keypoints are supervised as Gaussian-smoothed targets and decoded by local-maximum extraction. We adopt the loss-normalisation convention of that lineage but depart from its target representation, for reasons established in Sec. 4.6.

Learned fiducial detection. Deep ChArUco [5] is the closest prior work: a two-headed convolutional network predicting corner locations and identities on a coarse cell grid, with a separate refinement network. We adopt its two-stage decomposition and use it as our learned baseline. Conv-ChArT differs in the attention bottleneck, the full-resolution heatmap head, the geometric gate, and a substantially smaller parameter budget.

Attention in dense prediction and gating. Self-attention bottlenecks in U-Net-style encoders [8] are established in segmentation. We use a pre-norm transformer block with axial rotary position embedding [10, 11], chosen so that the encoding is relative and the exported graph remains free of operators unsupported by embedded inference runtimes. We adopt the additive attention gate of Attention U-Net [9] on the finest skip connection, with its pass-through initialisation.

3 Method

We first define the task and notation (Sec. 3.1), then describe the network (Sec. 3.2, overviewed in Fig. 1), its supervision (Sec. 3.3), the refinement and geometric stages that produce a pose or a refusal (Sec. 3.4), and the training configuration (Sec. 3.5).

3.1 Problem definition and notation

Given a grayscale frame $I \in \mathbb{R}^{H \times W}$ containing a 5×5 ChArUco board with $n = 16$ interior corners, we re-

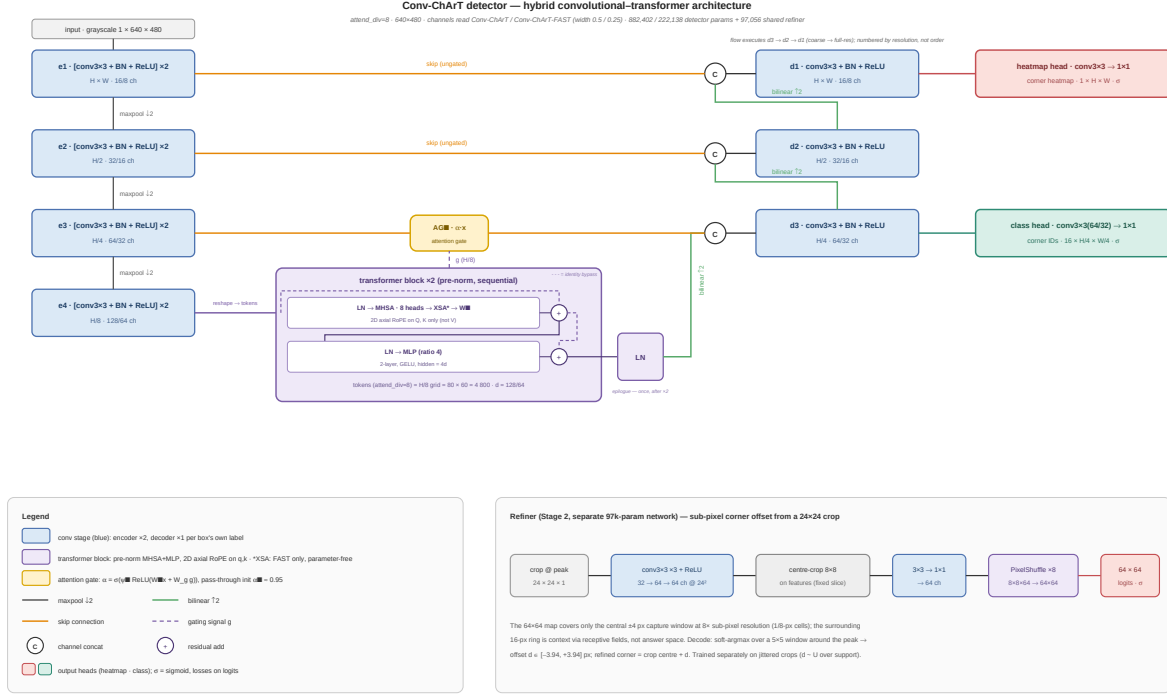


Figure 1: Conv-ChArT architecture. A grayscale frame enters a four-stage convolutional encoder; the deepest feature map is reshaped into an 80×60 token grid and passed through two pre-norm self-attention blocks with axial rotary position encoding, which supply the board-scale context that sets identity. A three-stage bilinear decoder recombines the attended features with encoder skips, the finest of which passes through an additive attention gate. Two heads emit a full-resolution corner heatmap and a 16-channel identity map at $H/4$. Peaks from the heatmap are cropped and refined sub-pixel by a separate 97,056-parameter network; identities are read from the class map at the coarse peak locations, tested for mutual consistency against a board homography, and either solved for pose or refused. The attention bottleneck holds 44.9% of the 882,402 parameters.

cover the image positions $\mathbf{x}_i \in \mathbb{R}^2$ and board indices $c_i \in \{1, \dots, n\} \cup \{\emptyset\}$ of the visible corners, and the camera pose $(\mathbf{R}, \mathbf{t}) \in SE(3)$. The system may decline to report a pose. Formally,

$$f(I) = (\{(\mathbf{x}_i, c_i)\}_{i=1}^K, (\mathbf{R}, \mathbf{t}) \cup \{\emptyset\}), \quad (1)$$

with K the number of detected corners, $c_i = \emptyset$ for a detection whose identity is not established, and the pose term $\emptyset$ when the correspondences fail the consistency test of Sec. 3.4. We write s for the apparent board square size in pixels; the operating envelope is $s \in [12, 128]$.

Refusal is an output, not a failure. A learned detector always produces a heatmap, so a pose solver downstream will always return something. Eq. 1 makes the refusal explicit and typed, which is what allows the error distribution to be bounded rather than merely centred (Sec. 4.3).

Half-pixel convention. Every resolution change uses $x' = (x+0.5)r-0.5$. Class-map readout uses bilinear sampling with `align_corners=False`, algebraically identical to mapping into cell space by $(x+0.5)/4-0.5$; dividing coordinates by 4 instead is biased by 0.375 cells, or 1.5 px at input resolution. Coordinates are float64 throughout, with no integer cast between the analytic corner and the

training target.

3.2 Architecture

Encoder. Four stages, each a pair of 3×3 Conv-BN-ReLU with stride-2 downsampling between stages, at width multiplier 0.5. The deepest stage is a plain convolutional stage. An earlier variant used a dilated cascade there to extend the receptive field; it is redundant once attention supplies board-scale context, and removing it costs nothing measurable while saving 25% of parameters (Sec. 4.6).

Attention bottleneck. The $H/8$ feature map is reshaped, without patch embedding, class token or absolute position embedding, into an 80×60 grid of 4800 tokens and passed through two pre-norm transformer blocks with 8 heads. Position is encoded by axial rotary embedding [10, 11] with wavelengths geometric over $[2.5, 2 \max(H', W')]$ cells. The longest wavelength exceeding twice the grid extent makes the joint phase code injective over all in-frame offsets, so a distant token pair cannot alias onto a near one. This bottleneck is the sole board-scale context mechanism, and is what makes identity recoverable at large s , where a corner's own neigh-

Module	Parameters	Share
Transformer blocks	396,544	44.9%
Encoder stage 4	221,696	25.1%
Decoder stage 3	110,720	12.5%
Encoder stage 3	55,552	6.3%
Class head	37,968	4.3%
Decoder stage 2	27,712	3.1%
Encoder stage 2	13,952	1.6%
Decoder stage 1	6,944	0.8%
Attention gate	6,209	0.7%
Encoder stage 1	2,512	0.3%
Heatmap head	2,337	0.3%
Detector total	882,402	
Refiner (separate network)	97,056	

Table 1: Parameter distribution of the reference model.

bourhood contains no marker.

Decoder and heads. Three bilinear-upsampling stages with skip concatenation. The finest skip passes through an additive attention gate [9] initialised to pass-through, gating only where the recovery pass of Sec. 3.4 can undo an incorrect veto. A full-resolution heatmap head and a 16-channel class map at $H/4$ terminate the network; final logit convolutions are bias-initialised to -2.19 following [6, 7]. We note this is not prior-matched to our task: $\sigma(-2.19) = 0.1006$ against a one-hot positive rate of $16/307,200 \approx 5.2 \times 10^{-5}$, so the initialisation is inherited rather than derived, and a matched value would be near -9.9 . We did not ablate it.

Table 1 gives the parameter distribution. We release two further tiers at width multipliers 0.375 and 0.25. The 502k tier is identical in topology to the reference; the 222k tier additionally enables exclusive self-attention (XSA) in the bottleneck, a parameter-free variant in which a token is excluded from its own attention distribution. At that width XSA measures -0.0040 px p95 and $+0.21$ pp identity against an otherwise identical control, both well inside that width’s reproducibility floor (Table 6); we adopt it because it is free, not because the difference is resolvable, and state it as such. This is why the 222k export carries 284 ONNX nodes against 263 for the other two tiers (Sec. 4.5).

3.3 Supervision

Let $\hat{p}$ denote a predicted probability and Y the target at the same location. The heatmap head is supervised by binary cross-entropy against a one-hot target, in which only the pixel nearest each visible corner is positive:

$$\mathcal{L}_{\text{hm}} = -\sum_{\mathbf{x}} \left[\mathbb{I}[Y=1] \log \hat{p} + \mathbb{I}[Y<1] \log(1 - \hat{p}) \right]. \quad (2)$$

The class head is supervised by the penalty-reduced focal loss of [6] against a Gaussian target of width σ_{cls} :

$$\mathcal{L}_{\text{cls}} = -\sum_{\mathbf{x}} \begin{cases} (1 - \hat{p})^\alpha \log \hat{p}, & Y = 1, \\ (1 - Y)^\beta \hat{p}^\alpha \log(1 - \hat{p}), & Y < 1, \end{cases} \quad (3)$$

with $\alpha = 2$, $\beta = 4$. Both are computed in logit space. The naive form is unusable in bf16 not through underflow, since bf16 carries fp32’s 8-bit exponent, but through mantissa precision: with 8 mantissa bits, $1 - \hat{p}$ rounds to zero once $\hat{p} \gtrsim 1 - 2^{-9}$, which the forced target peaks drive it to. The total objective is

$$\mathcal{L} = \frac{1}{N} \left[\mathcal{L}_{\text{hm}} + \lambda_{\text{cls}} \mathcal{L}_{\text{cls}} \right], \quad N = \max\left(\sum_b N_b, 1\right), \quad (4)$$

where N is the batch-total count of visible corners, shared by both heads and clamped at one so that all-negative batches require no special case.

Eq. 3 reduces to Eq. 2 when $\alpha = 0$ and $Y \in \{0, 1\}$. Using the one-hot form on the heatmap departs from the convention of [6, 7]; it improves localisation because the graded target’s proximity discount makes displaced probability mass nearly free, which we isolate in Sec. 4.6. The two smaller tiers use Eq. 2 for both heads; λ_{cls} is 2.0 at the reference and 222k widths and 1.5 at 502k.

3.4 Refinement, correspondence and pose

Sub-pixel refinement. Peaks are decoded from the heatmap by 3×3 max-pool equality, deduplicated within 2px, and capped at 64. A separate 97,056-parameter network takes a 24×24 crop centred on each peak and predicts a 64×64 logit map, decoded by a probability-weighted centroid over a 5×5 window around the argmax. The refiner never upsamples its input; resolution is gained by pixel-shuffling learned channels, which keeps the exported graph free of resize operators.

Confidence gate. Refinement is applied only where the refiner’s own peak confidence exceeds 0.3; below that the coarse peak is retained. The refiner is worse than the coarse peak on 16.3% of crops, predominantly those without recoverable corner structure, and no amount of training recovers a sub-pixel position from a crop that does not contain the corner. We gate on the refiner’s own output rather than on an image statistic so the decision uses the same evidence the refinement does; the gate is ablated in Sec. 4.6 (*Geometric gating*).

Identity readout. Identity is read from the class map at the *coarse* peak positions. The class map is at $H/4$, so sub-pixel position carries no identity information; more importantly, the gate and recovery pass below both reassign identity on the strength of positions, so feeding them refined coordinates would let the refiner rewrite identity indirectly. Decoupling makes the refined and unrefined arms bit-identical in identity by construction.

Input resolution	640×480 , grayscale
Steps	100,000 (detector), 10,000 (refiner)
Batch	16×2 accumulation (effective 32)
	512 (refiner)
Optimiser	AdamW, $\beta = (0.9, 0.999)$
Peak / floor LR	3×10^{-4} / 3×10^{-6}
Schedule	cosine, 1000 warmup steps
Weight decay	10^{-4} , excl. biases and norms
Gradient clip	5.0
EMA decay	0.999 (evaluated)
Precision	bf16 autocast, fp32 master weights

Table 2: Training configuration, shared by all three tiers.

Lattice-consistency gate. A homography is fitted by RANSAC from the identified corners to the canonical board lattice. Corners inconsistent with the fit are demoted; a recovery pass then re-projects all board corners through the homography to assign identities the class head missed. Degenerate configurations, namely fewer than four identified corners, collinear sets, and exact-four fits that are vacuous by construction, yield typed refusals rather than a pose. Pose is solved by IPPE [12] on undistorted coordinates, with an ambiguity flag raised when the two IPPE solutions’ reprojection errors are within a factor of 1.5. The flag fires on 4.2/4.1/4.6% of solved frames across our three tiers and is reported rather than acted on: it is a second, softer refusal channel that a consumer may use to discount a pose without discarding it.

3.5 Implementation details

Table 2 gives the training configuration; all three tiers share it. Training runs in PyTorch 2.8 on a single RTX 5090 and takes 8.4 h for the 882k and 502k tiers and 9.4 h for the 222k tier; the refiner is trained separately (1.7 h). Training data is generated on the fly. Appendix A gives the loader configuration, the full architecture specification, and the determinism and provenance machinery.

4 Experiments

4.1 Data, metrics and baselines

Training data. No annotated corpus exists at the scale and degradation severity needed to characterise behaviour across the operating envelope, and manual sub-pixel annotation of chessboard saddle points is infeasible at the precision the task demands. We train entirely on synthetic composites with analytically exact labels: a rendered board is warped by a sampled homography onto a natural-image background, alpha-composited, occluded, and photometrically degraded, with corner positions computed through the same homography rather than from pixels. Apparent square size is log-uniform over $[12, 128]$ px and perspective is sampled as a physical

Tier	Params	p95 (px)	Median (px)	M-04 (%)	Tail (%)	Far-oct. rec. (%)
222k	222,138	0.734	0.413	99.09	0.019	87.5
502k	502,322	0.689	0.408	99.41	0.015	90.5
882k	882,402	0.673	0.406	99.53	0.005	91.7
<i>floor, 222k</i>		0.023		6.18	0.010	
<i>floor, 882k</i>		0.002		0.07	0.005	

Table 3: Corner detection, full 10,000-sample validation at 100,000 steps, with the seed-to-seed floor of Table 6 repeated at the ladder’s two endpoints. Figures are quoted to one digit past the 882k floor. Far-octave recall is M-02 in the $s = 64$ –128 bin.

tilt up to 60° . Appendix B details the photometric model and labelling policy.

Metrics. **M-01** is the localisation error of matched corners under greedy nearest-neighbour matching within 8 px, reported as median and 95th percentile. **Tail** > 4 px is the fraction of matched corners outside the refiner’s capture radius, our operational gate. **M-02** is match ratio by scale octave. **M-04** is corner identification accuracy, conditioned on detection; a model detecting fewer, easier corners is scored on an easier set, so we report recall alongside it throughout. **M-05/M-06** are pose rotation (geodesic) and translation error, the latter in board squares.

Baseline provenance. The classical baseline is the OpenCV ChArUco pipeline at default detector parameters. The learned baseline is Deep ChArUco [5]; we did not copy its published numbers, which were obtained on a different distribution. We initialised from the public reference checkpoint, reinitialised the identity head for our board dictionary, and fine-tuned on our training distribution under the same early-stopping criterion applied to our own arms, terminating at 62,458 steps on convergence. Its corners pass through the identical correspondence-gating and pose stage as ours, so the comparison isolates corner quality rather than pose-solver implementation. Its refinement network is not run, which we state rather than hide; its 320×240 input against our 640×480 is a further asymmetry.

4.2 Corner detection

Table 3 reports the three released tiers. Across a $4 \times$ parameter range, in-envelope identity spans 0.44 pp and median localisation 0.0065 px; both are inside the 222k floor and so are not resolvable at the ladder’s lower end. Of the three separations that do exceed it, the 95th percentile ($+0.060$ px, $2.6 \times$ the 222k floor) and the tail ($4.0 \times$, $1.4 \times$ the floor) survive, and far-octave recall (-4.2 pp) is the largest. The tiers therefore separate on tail behaviour and far-range recall, not on central accuracy. Note that the 222k tier also differs architecturally (Sec. 3.2), so this row-to-row comparison is between released *configurations*

rather than a clean capacity sweep.

4.3 Pose estimation

Every Conv-ChArT tier attains lower rotation and translation error than Deep ChArUco at a fraction of its parameter count (Table 4), and our smallest pipeline, 7.0 $\times$ smaller, still surpasses it on every accuracy metric.

Solve rate is not comparable across methods. Deep ChArUco’s 93.3% lies between our 502k and 882k tiers, but its 95th-percentile rotation error is 89.36 $^\circ$ against our 1.17 $^\circ$, and its mean error is 11.6 $\times$ its own median, against 4.1 $\times$ for our reference model: a distribution dominated by catastrophic solves. It fails by emitting a confidently wrong pose where our gate refuses. The classical pipeline exhibits the opposite failure mode, competitive median accuracy on what it solves but abstention on 45.6% of frames. Our *unrefined* coarse output (0.374–0.382 $^\circ$ median) already exceeds the learned baseline’s accuracy.

Capacity purchases the tail, not the median. Median rotation error is flat across our ladder (0.1269/0.1295/0.1318 $^\circ$), but the mean falls monotonically (0.6855 $\rightarrow$ 0.5759 $\rightarrow$ 0.5368 $^\circ$) and mean translation error falls 46% from the smallest to the largest tier.

4.4 Robustness

Figure 2 and Table 5 report twenty degradation factors on identical frames. The classical pipeline’s identity column is $\sim$ 100% by construction, since it reports only corners it has already decoded; its recall is the informative column.

Capacity buys identity under degradation. Board occlusion is the clearest case: identity accuracy spans 40.2% $\rightarrow$ 56.8% $\rightarrow$ 71.2% across our tiers, a 31 pp separation on frames where clean validation separates the same models by 0.44 pp. Clean-set metrics substantially understate what additional capacity provides.

A deployment caution. The smallest tier collapses to 5.6% recall at the most severe darkness step, against 37.3% and 45.8% for the larger tiers. This is a cliff rather than a graceful decline, and it is invisible in clean-validation numbers.

4.5 Efficiency and deployment

All four networks export to ONNX opset 17 with no operator outside the standard embedded-runtime set (detectors 284/263/263 nodes at 2.05/3.70/5.74 MB; refiner 20 nodes at 0.37 MB) and agree with the reference implementation to $\sim 10^{-4}$ in logit space. Half-precision behaviour separates by stage: applying fp16 to each network independently displaces 3 corners (0.062%) beyond 0.05 px for the detector alone and 41 (0.846%) for the refiner alone, additively recovering the 44 (0.908%) observed when both are quantised. The refiner accounts for the entire sub-pixel displacement tail and none of the identity change.

Retaining the 97,056-parameter refiner at full precision while quantising the detector reduces the displaced fraction to 0.008–0.067%; uniform quantisation of both networks fails our acceptance criterion at every tier. That criterion, fixed before the measurement, is that fewer than 0.1% of corners may be displaced by more than 0.05 px and identity may change on fewer than 0.1%. We note that “fp32” convolutions execute in TF32 by default on current hardware, whose 10-bit mantissa matches fp16’s; a comparison against an undisabled TF32 reference understates the effect.

4.6 Ablation studies

Every arm differs from its control by exactly one configuration key, verified programmatically before launch, and is trained to a matched budget on an identical data stream. Deltas are quoted against the control of the same width and budget, and against that width’s reproducibility floor. Appendix C reports the remaining studies.

Reproducibility floor. We train repeated seeds of an identical configuration and take the range of the final 10,000-sample validation (Table 6). The floor is a joint property of width and metric: the p95 range at 882k is 11.6 $\times$ tighter than at 222k, and the refiner resolves median error 55 $\times$ more finely than its own p95. Two consequences follow. First, at 222k an identity range of 6.18 pp exceeds every effect we measure at that width, so we draw no conclusions there. Second, mid-training ranges are far larger than final ones: on the in-loop 2000-sample validation the 502k four-seed identity range is 35.39 pp at step 7500, against 0.09 pp on the full set at step 35,000. Intermediate validations therefore measure which seed converged first, not which configuration is better.

Attention bottleneck. *W/o attention* replaces the two transformer blocks with an equal-depth convolutional stack, leaving encoder, decoder and heads unchanged. At 4.7M parameters and matched budget it gives 98.25% identity against 98.80% for the control, and the deficit concentrates at large s , where a corner’s own neighbourhood contains no marker and identity must come from board-scale context. Within the bottleneck, halving the head count, halving the attention resolution (which adds 607k parameters), removing one of the two blocks, and changing the rotary wavelength anchor all lie within the 0.0020 px / 0.07 pp floor for this width (Table 10). The bottleneck is therefore necessary but already at its smallest useful configuration, which is what permits the parameter budget of Table 1.

Supervision signal. *W/o one-hot supervision* restores the convention of [6]: a Gaussian heatmap target of width σ_{hm} under Eq. 3. Sweeping σ_{hm} to the one-hot limit at the reference width and matched budget gives 0.7781, 0.7352, 0.7335 and **0.6970** px p95 at $\sigma_{\text{hm}} = 1.0, 0.5, 0.25$ and one-hot respectively, with identity varying by 0.14 pp across the Gaussian rungs, twice this width’s floor but an order of magnitude below the localisation effect, and the

Method	Params	Solve (%)	Rotation error (°)			Translation (squares)		
			Median	Mean	p95	Median	Mean	p95
Classical (OpenCV)	—	54.4	0.1602	0.8690	2.3487	0.0200	0.0654	0.2236
Deep ChArUco [5]	2,245,992	93.3	0.9131	10.6234	89.3603	0.0770	1.2636	7.2695
Conv-ChArT 222k	319,194	90.0	0.1269	0.6855	1.2261	0.0099	0.0691	0.1523
Conv-ChArT 502k	599,378	92.3	0.1295	0.5759	1.2297	0.0103	0.0604	0.1509
Conv-ChArT 882k	979,458	93.6	0.1318	0.5368	1.1737	0.0104	0.0376	0.1282
Conv-ChArT 882k (coarse)	882,402	93.6	0.3744	1.1176	2.3424	0.0289	0.0739	0.2698

Table 4: Pose estimation on 1000 images held out from the training generator, i.e. in-distribution generalisation. Parameter counts are full pipelines: baseline 1,244,572 (detector) + 1,001,420 (refinement network); ours 222,138/502,322/882,402 (detector) + 97,056 (refiner). The baseline’s refinement network is not run, so its localisation is coarse-stage only; the final row gives our coarse-stage output, which is the like-for-like comparison and still improves median rotation error $2.4\times$ over the baseline. All arms’ corners pass through the identical correspondence-gating and PnP stage. Our smallest pipeline is $7.0\times$ smaller than the learned baseline and surpasses it on every accuracy metric; our reference model is $2.3\times$ smaller and reduces median rotation error $6.9\times$ and 95th-percentile rotation error $76\times$.

Factor	222k	502k	882k	DC	Cls.
Distance $s=128$	91.7	92.6	92.8	89.6	18.6
Darkness	5.6 [†]	37.3	45.8	0.2	0.0
Sensor noise	16.5	15.4	16.6	13.8	2.6
Motion blur	63.6	75.3	72.8	76.8	12.5
Ink contrast	54.9	63.6	70.3	58.5	11.3
Object occl. $\times 4$	92.8	94.2	96.0	87.5	32.8
Board occl. 50%	82.3	87.1	85.9	73.1	9.6
Tilt 60°	91.5	91.9	92.7	90.0	51.6

Table 5: Detection recall (%) at the most severe step of each factor, for eight of the twenty factors measured; the full set is listed in Appendix C. Bold marks the best value in a row; [†] marks a value discussed in the text for being anomalous rather than best. DC = Deep ChArUco, Cls. = classical.

tail improving $8.8\times$ at the limit. The mechanism is that the graded target enters the loss only through the $(1-Y)^\beta$ factor on the negative branch: at $\sigma_{\text{hm}} = 2$, $\beta = 4$, a pixel one pixel from a corner has $Y \approx 0.8825$ and therefore weight 1.9×10^{-4} , so displaced probability mass is nearly free. Measured on identical frames ($n = 830$), peaks under Gaussian supervision have RMS radius 1.0028 px with 26.1% of local mass on the peak pixel, against 0.2665 px and 93.5% under one-hot supervision; since localisation decodes an argmax, a broader peak is directly a noisier estimate. Setting $\beta = 0$ isolates the two terms of Eq. 3: it recovers 76.6% of the localisation gain but only 8.5% of the identity gain, so the proximity discount governs localisation and the easy-example modulation governs identity, which is why the two heads are supervised differently. Full ladders are in Appendix C.

Geometric gating. *W/o the confidence gate* applies the refiner to every crop. This raises the 95th-percentile refinement error from 0.761 px to 3.011 px while moving the median by 0.0012 px: the refiner is worse than the coarse peak on 16.3% of crops, of which 55% are featureless (intensity standard deviation < 15 , against 20%

Width	Seeds	M-04 range	p95 range
222,138	4	6.18 pp	0.0231 px
502,322	4	0.09 pp	0.0061 px
882,402	3	0.07 pp	0.0020 px
Refiner (97,056)	3	med. 0.0017 px	p95 0.0940 px

Table 6: Seed-to-seed range of an identical configuration, on the full 10,000-sample validation at the 35,000-step ablation budget. Applied to the released 100,000-step models of Table 3 it is indicative rather than exact, since reproducibility is itself a function of budget.

of the remainder), and the gate fires on 13.8% of crops and is correct 69.5% of the time. *W/o the lattice gate* passes all identified corners to PnP without the consistency test, which is the configuration the learned baseline is effectively in; Table 4 shows the consequence, a 95th-percentile rotation error of 89.36° against 1.17°. Neither gate improves the median: both truncate the tail, which is the property that matters when poses are consumed without inspection.

5 Discussion

Limitations. Training is entirely synthetic; we have not evaluated on annotated real imagery, and the photometric model, while calibrated to a specific sensor class, is a model. The pose evaluation sets omit the segmentation-derived object occluders used in training, so object occlusion is characterised only on its own robustness axis. The classical baseline is evaluated at default parameters, and a practitioner-tuned configuration would likely improve its recall.

Uncertainty. Our per-corner scale estimate is a useful ordering signal but is not a variance, and the pose covariance is approximately $6\times$ over-confident. We report this rather than applying a fitted inflation factor, which would

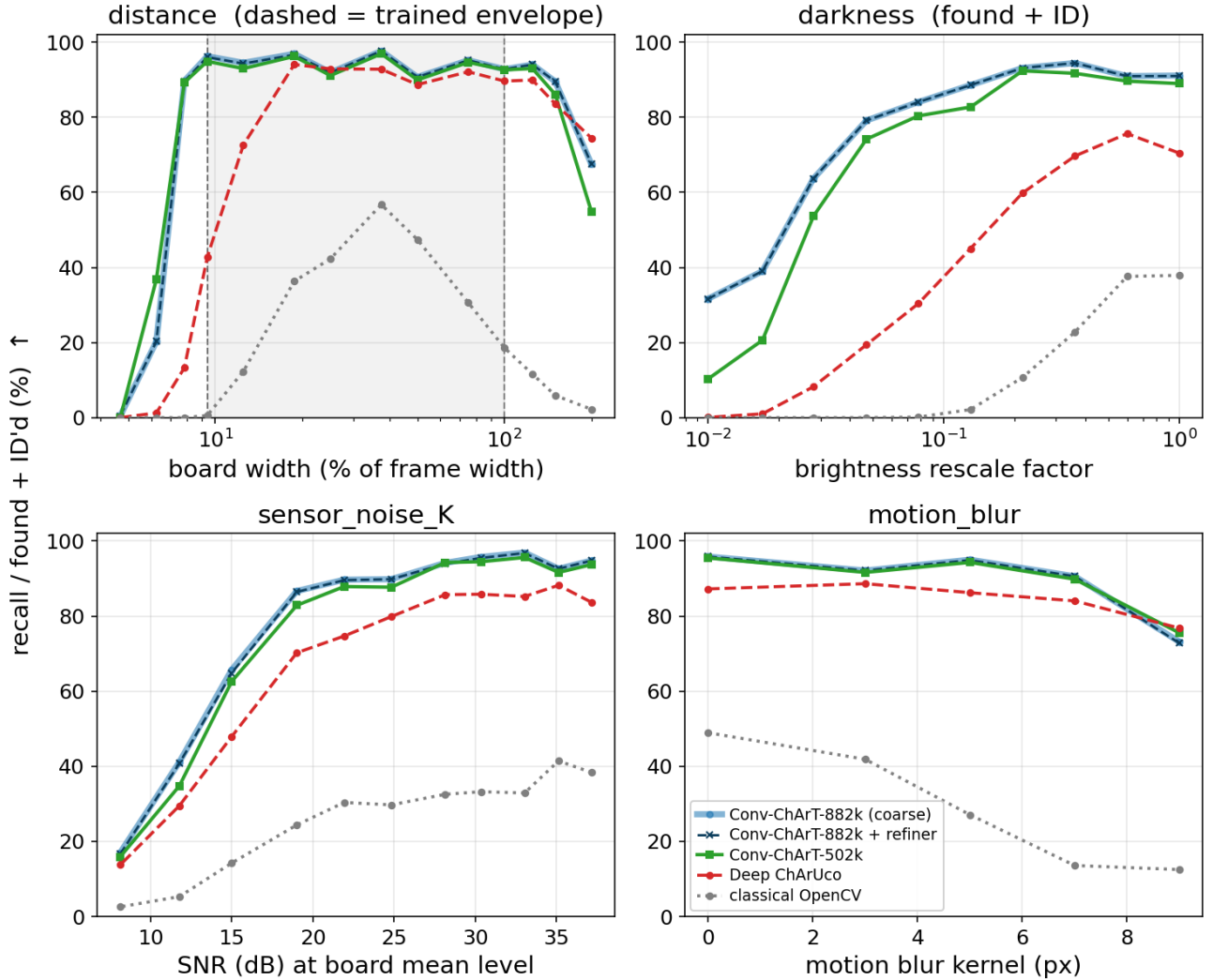


Figure 2: Detection recall across four degradation axes, on identical frames. The classical pipeline degrades first on every axis; Conv-ChArT retains recall furthest into distance and low light.

conceal the mis-specification rather than correct it.

Scope of the null results. The structural variants in Appendix C are established within our operating envelope, board geometry and data distribution, at the sample sizes stated. They are not claims about attention in dense prediction generally.

6 Conclusion

We presented Conv-ChArT, a convolutional–transformer hybrid for ChArUco corner detection and pose estimation that establishes a new accuracy–efficiency operating point for this task. Our full pipeline is $2.3\times$ smaller than the prior learned state of the art while reducing median rotation error $6.9\times$, and a $7.0\times$ smaller variant still surpasses it on every accuracy metric. Two design choices account for this. Abstention is geometric and explicit, bounding the error distribution rather than merely lowering its cen-

tre; and the supervision departs from the dense-keypoint convention in a way that measurably sharpens the decoded peak. Reporting every ablation against a measured reproducibility floor is what allowed us to size the network by evidence rather than convention, and we recommend the practice wherever reported effects are small relative to training stochasticity.

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A Extended implementation details

Architecture. Encoder widths at multiplier 0.5 are 16/32/64/128 channels across stages e1–e4, each a pair of 3×3 Conv–BN–ReLU with stride-2 downsampling between stages. The bottleneck operates on the $H/8$ map at embedding dimension 256 with 8 heads, MLP ratio 4, LayerNorm $\varepsilon = 10^{-6}$, no dropout, and qkv bias enabled. Rotary embedding uses 8 frequencies per axis over 16 dimensions per axis, with fp32 phase buffers. The decoder uses bilinear upsampling followed by 3×3 convolution, with skip concatenation at each rung. The refiner is 3×3 conv $1 \rightarrow 32$, $32 \rightarrow 64$, $64 \rightarrow 64$ at 24^2 , a centre crop to 8×8 , 3×3 $64 \rightarrow 64$, 1×1 $64 \rightarrow 64$ logits, then PixelShuffle(8) to 64×64 .

Training. Batch 16 with accumulation 2; AdamW $\beta = (0.9, 0.999)$; peak learning rate 3×10^{-4} , floor 3×10^{-6} , cosine schedule anchored to the total step budget, 1000 warmup steps; weight decay 10^{-4} applied to neither biases nor normalisation parameters; gradient clipping at 5.0; EMA decay 0.999, evaluated in place of the live weights. Mixed precision is bf16 with fp32 master weights. The data loader uses a spawn start method, 12 workers, prefetch factor 4, and single-threaded OpenCV per worker. Validation runs every 2500 steps on a fixed 2000-sample subset and every 25,000 steps on the full 10,000-sample set; all reported numbers are from the full set.

Determinism and provenance. Every sampler takes an explicit random generator; the codebase contains no global random-number use, enforced by a static scan in the test suite. A content hash of the generator sources and of the configuration subset that shapes their output is recorded with every checkpoint, and training refuses to start if either has drifted since the last passing data audit. Reported detection numbers are single runs; the seed spread of an identical configuration is quantified in Table 6 and should be read as the uncertainty on any single number.

B Synthetic data generation

Photometric model. Applied in a pinned order, each stage independently gated by its own probability: illumination, occlusion, specular lobe, refractive droplets, vignette, near-infrared ink-contrast jitter, blur family (motion and defocus), Poissonian–Gaussian sensor noise [13], fixed-pattern noise, speckle, and contrast. The model targets an industrial global-shutter monochrome camera with a narrow-band near-infrared filter and no consumer image-signal-processor stages.

Reported signal-to-noise ratio is that of the delivered frame. Noise is drawn per colour channel before grayscale conversion, and the luminance-weighted average of three independent draws attenuates the delivered standard deviation by $\sqrt{0.114^2 + 0.587^2 + 0.299^2} = 0.6686$, measured 0.6676–0.6789 over $\sigma \in [3, 12]$ DN. The delivered frame is therefore 3.50 dB quieter than the per-channel Poissonian–Gaussian formula implies, and we report the delivered figure.

Labelling policy. Corner visibility is geometric only: a corner is visible if it lies in frame and outside every occluder. Photometric degradation never changes a label, which keeps the label a function of geometry alone and prevents the network being supervised to hallucinate corners it cannot see. A corner whose crop is illegible retains its true index, so the class loss on illegible corners is reducible only through the attention pathway.

C Further ablations

All arms are trained to a matched budget with exactly one configuration key changed, and are compared against the control of the same width. Values are p95 (px), M-04 (%) and tail (%).

The twenty degradation factors. Each factor is swept over a ladder of severity steps on frames that are otherwise bit-identical across methods and steps, so that only the swept parameter varies. The factors are: *geometric* — distance ($s = 12$ to 128 px), distance-extrapolated ($s = 6$ to 256, outside the trained envelope), rotation, tilt; *photometric* — brightness, darkness, contrast, ink contrast, vignette, specular, droplets; *sensor* — sensor noise (gain K), motion blur, defocus blur; *occlusion* — board occlusion (percentage of board area), object occlusion (segmentation-derived cutouts), rectangular occlusion; and *differencing* — differencing ratio, differencing ambient level, differencing ghosting. Table 5 reports eight of these, chosen to span all five groups; the remaining twelve follow the same qualitative pattern and are included in the released result set. The severity ladders are defined in the released sweep configuration rather than reproduced here.

Heatmap target and loss form. Table 7 gives the target-width ladder at the reference width and the loss decomposition at 4.7M parameters.

Per-head supervision. The two heads need not share a loss form (Table 8). At 882k, one-hot BCE on the heatmap improves p95 under either class-head setting, and focal on the class head gives the best p95 and tail of the

(a) 882,402 params, 35k steps				(b) 4,698,034 params, 50k steps			
Target	p95	M-04	Tail	Supervision	p95	M-04	Tail
Gaussian $\sigma_{\text{hm}}=1.0$	0.7781	98.90	0.1006	$\sigma_{\text{hm}}=2.0$, focal $\beta=4$	0.8149	98.80	0.1221
Gaussian $\sigma_{\text{hm}}=0.5$	0.7352	98.86	0.0849	$\sigma_{\text{hm}}=1.0$, focal $\beta=4$	0.7430	99.06	0.0687
Gaussian $\sigma_{\text{hm}}=0.25$	0.7335	99.00	0.0803	$\sigma_{\text{hm}}=0.25$, focal $\beta=4$	0.7016	99.05	0.0392
One-hot	0.6970	99.29	0.0114	$\sigma_{\text{hm}}=2.0$, $\beta=0$	0.7032	98.87	0.0463
				One-hot, BCE	0.6690	99.62	0.0056

Table 7: Heatmap target width and loss decomposition.

four. At 222k the same combination collapses identity to 58.54%, far outside that width’s 6.18 pp range, which is why it is not carried to that tier.

Heatmap loss	Class loss	p95	M-04	Tail
<i>882,402 params, 35k steps</i>				
focal (Gaussian)	focal (Gaussian)	0.7352	98.86	0.0849
focal (Gaussian)	BCE (one-hot)	0.7572	99.00	0.0931
BCE (one-hot)	focal (Gaussian)	0.6874	98.87	0.0081
BCE (one-hot)	BCE (one-hot)	0.6970	99.29	0.0114
<i>222,138 params, 35k steps</i>				
focal (Gaussian)	focal (Gaussian)	0.8096	94.87	0.0952
focal (Gaussian)	BCE (one-hot)	0.8222	96.71	0.1265
BCE (one-hot)	focal (Gaussian)	0.7430	58.54 [†]	0.0182
BCE (one-hot)	BCE (one-hot)	0.7611	94.76	0.0218

Table 8: Per-head loss form, crossed at two widths, 35k steps. Bold marks the best value in a column within each width block; † marks the identity collapse discussed in the text.

Head balance. λ_{cls} scales the class-head term of Eq. 4. At 882k it is a trade at every setting, buying identity at roughly 0.06 px of p95 per point; at 502k it is free within the measured range, which is why the released tiers differ (Table 9).

Width	λ_{cls}	p95	M-04	Tail
882,402	1.5	0.7021	99.34	0.0130
882,402	2.0	0.7059	99.40	0.0098
882,402	4.0	0.7143	99.51	0.0115
502,322	1.5	0.7122	98.98	0.0124
502,322	2.0	0.7208	99.09	0.0140
502,322	3.0	0.7234	99.12	0.0165
502,322	4.0	0.7262	99.29	0.0124
222,138	0.5	0.7318	91.96	0.0192
222,138	2.0	0.7580	97.80	0.0168
222,138	4.0	0.7703	97.78	0.0168

Table 9: Class-head weight, 35k steps.

Class-target and refiner-target width. In contrast to σ_{hm} , the class-target width is null in both directions: at 502k, $\sigma_{\text{cls}} = 0.5$ gives 0.7110 / 98.87 and $\sigma_{\text{cls}} = 2.0$ gives 0.7104 / 98.80, both within that width’s range of the 0.7139 / 98.80 control. The class map is at $H/4$, so one cell spans four input pixels and sub-pixel structure in its target is not recoverable. The refiner-target width is likewise null (median 0.0937 against 0.0969 px at $\sigma_{\text{ref}} = 0.75$ and 1.5), against a median range of 0.0017 px but a p95 range of 0.0940 px that cannot arbitrate the p95 column.

Attention bottleneck internals. Table 10 varies the bottleneck against a matched control at the reference width. All four variants lie within the 0.0020 px / 0.07 pp floor, so we retain the smallest configuration.

Skip gating. *W/o gate* removes the additive attention gate on the $H/4$ skip, passing the encoder features through unmodified. This changes p95 by -0.0011 at 882k (0.6959 against 0.6970) and $+0.0018$ at 502k, both within range.

Variant (882,402 params, 35k steps)	p95	M-04	Tail
Control (8 heads, $H/8$, 2 blocks)	0.7352	98.86	0.0849
4 heads	0.7359	98.77	0.0746
Attention at $H/16$ (+607k params)	0.7398	98.96	0.0586
1 block	0.7398	98.51	0.0803
RoPE anchor $\lambda_{\min}=5$ cells	0.7400	98.89	0.0794

Table 10: Attention bottleneck variants.

Adding a second gate on the $H/8$ skip gives 0.7063 / 98.89 at 502k against 0.7139 / 98.80 for the control, i.e. $1.2\times$ and $1.0\times$ the respective ranges. Gating is retained on the $H/4$ skip only, at 6209 parameters.

Encoder dilation, cross-scale attention and width. At 4.7M parameters and 50k steps, removing the dilated cascade from the deepest encoder stage gives 0.8072 / 98.77 against the 0.8149 / 98.80 control while removing 25% of parameters; adding cross-scale attention gives 0.8148 / 98.82; disabling gates gives 0.8134 / 98.88. Halving the width gives 0.8605 / 98.67, the only structural change with an effect clearly outside the range, and it is a degradation.

Optimisation. Raising the peak learning rate from 3×10^{-4} to 6×10^{-4} gives 0.7100 / 99.30 against 0.7238 / 98.10 at 1.5×10^{-4} , at 502k and 35k steps. We retain 3×10^{-4} for the released models, whose longer schedule alters the comparison.

Composite configurations. Levers do not compose additively. Combining the two-gate topology with $\lambda_{\text{cls}} = 2$ at 222k gives 0.7624 / 97.25, worse on both axes than $\lambda_{\text{cls}} = 2$ alone (0.7580 / 97.80). At 882k, one-hot heatmap with focal class head and $\lambda_{\text{cls}} = 2$ gives 0.6894 / 99.15 with a tail of 0.0065%, the best tail measured at that width and the configuration we release.

Training budget. Our 502,322-parameter model at 100,000 steps (0.6885 / 99.41) outperforms the 882,402-parameter model at the 35,000-step ablation budget (0.6894 / 99.15) on both axes. Ablation deltas are valid at matched budget; absolute ablation values are budget-limited and should not be compared with the released models of Table 3.